



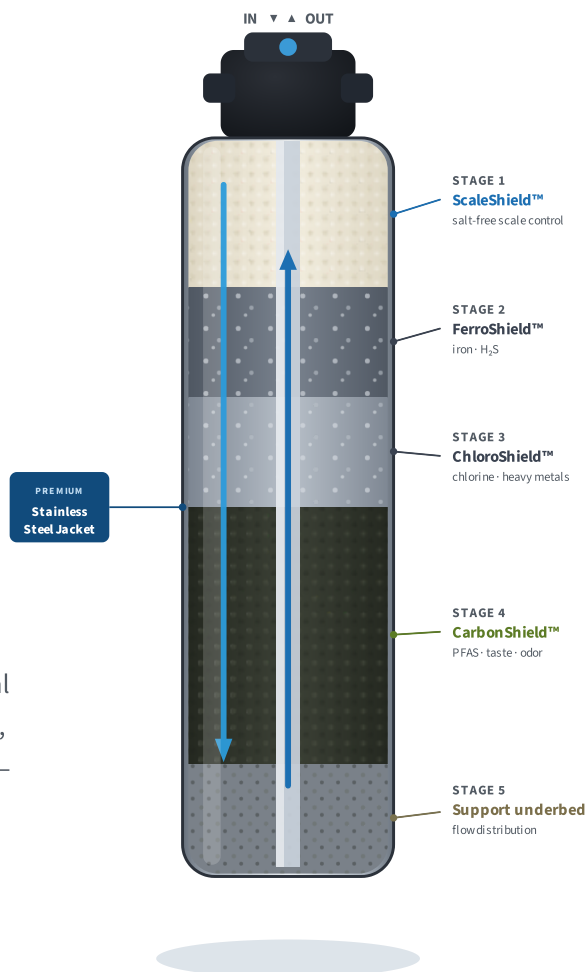
ONE GREEN FILTER | 5-STAGE WHOLE-HOME FILTER

PFAS + SCALE DEFENDER

— One Green Defender™ Series

One revolutionary tank with five proprietary treatment layers, housed in a **premium full stainless-steel jacket** engineered to last. The **PFAS + Scale Defender** reduces **up to 98% of PFAS**, inhibits bacteria and algae from growing inside the system, and prevents limescale — completely **salt-free**, with no brine and no sodium added to your water.

Water enters the top and passes down through our **ScaleShield™** crystal scale-prevention media, **FerroShield™** and **ChloroShield™** redox layers, a deep **CarbonShield™** catalytic bed, and a graded support underbed — then returns up the center riser, fully treated.



Up to 98%
PFAS / PFOA reduction



100% Salt-Free
No brine tank, no sodium



Bacteriostatic
Stops bacteria & algae growth



Minerals Kept
Healthy Ca & Mg preserved



FIVE POWERFUL STAGES. ONE SMART TANK.

The PFAS + Scale Defender is engineered around a **premium full stainless-steel jacketed mineral tank** with five proprietary OneGreenFilter media layers stacked from inlet to underbed. Water enters the top and passes **straight down through all five stages** in sequence, then returns up the center riser, fully treated, to your home.

1

SCALESHIELD™ CRYSTAL — *Salt-Free Scale Prevention*

A proprietary nucleation-crystallization media that transforms dissolved calcium & magnesium bicarbonate into stable, harmless **micro-crystals** that can't stick to pipes, heaters or fixtures — preventing new scale and even removing existing scale, with no salt, no regeneration and no sodium added.

2

FERROSHIELD™ PRO REDOX — *Iron, Hydrogen Sulfide & Bacteriostatic*

A proprietary high-purity copper-zinc redox media. Converts dissolved **ferrous iron** into filterable ferric hydroxide and eliminates **rotten-egg sulfur (H₂S)**. Its electrolytic field plus trace hydroxyl radicals **inhibit bacteria, algae and fungi** from growing inside the bed.

3

CHLOROSHIELD™ PRO REDOX — *Chlorine & Heavy Metals*

Removes **up to 99% of free chlorine** by electrochemically converting it to harmless chloride ions, plus **up to 98% of dissolved heavy metals** — lead, mercury, copper, nickel and chromium. Also adds to the bacteriostatic protection of the bed.

4

CARBONSHIELD™ CATALYTIC — *PFAS, Taste & Odor*

A proprietary metal-oxide and iron-enhanced catalytic coconut-shell carbon. Its **positively charged surface** breaks down the long carbon-fluorine chains of perfluorinated compounds, reducing **up to 98% of PFAS / PFOA**. High-surface-area carbon (2,000–2,500 m²/g) also captures chlorine, tastes, odors and sediment to **≤ 1 µm**.

5

SUPPORT UNDERBED — *Flow Distribution & Media Support*

A graded support underbed evenly distributes service and backwash flow across the cross-section of the tank, supports the four treatment layers above, and keeps media fines out of the riser and distributor.

FerroShield™ and ChloroShield™ redox media are bacteriostatic — they inhibit microbial growth within the media bed and are not a disinfection system. Pre-treatment may be required for very high iron, manganese or hydrogen sulfide. PFAS reduction depends on empty-bed contact time (EBCT ≥ 90 sec) and influent water chemistry.

WHY IT MATTERS

CLEANER, SAFER, SCALE-FREE WATER



PFAS PROTECTION

Targets the "forever chemicals" most carbon filters leave behind — up to 98% reduction.



SALT-FREE SOFTENING

All the scale protection of a softener with no salt to buy, haul, or store — and no brine drain line.



HEALTHY MINERALS

Keeps beneficial calcium & magnesium in your water — and adds zero sodium.



ECO-FRIENDLY

No salt discharge and no regeneration wastewater on the scale-prevention bed — greener for your home and your area.



WHOLE-HOME

Point-of-entry protection for every tap, shower, and appliance in the house.



BACTERIOSTATIC

Our proprietary redox media inhibits bacteria, algae & fungi from growing inside the tank — fresher water and longer media life.

DEFENDER VS. TRADITIONAL SALT SOFTENER

	Traditional Salt Softener	PFAS + Scale Defender
Reduces PFAS / forever chemicals	X	✓ up to 98%
Reduces chlorine, taste & odor	X	✓
Inhibits bacteria, algae & fungi in bed	X	✓ (redox)
Reduces iron, H ₂ S & heavy metals	X	✓
Prevents & removes limescale	✓	✓
Adds sodium to your water	Yes	None
Requires salt purchase & refills	Yes	No
Preserves healthy Ca & Mg minerals	X	✓



BATHROOM

Softer skin & hair, no soap scum.



KITCHEN

Spot-free glassware & dishes.



LAUNDRY

Brighter, longer-lasting fabrics.



APPLIANCES

Scale-free heaters last longer.

TECHNICAL DATA



RESIDENTIAL SYSTEM SIZING

Model	OGD-8	OGD-11	OGD-14
Stage 1 — ScaleShield™ Crystal	1.0 ft ³	1.5 ft ³	2.0 ft ³
Stage 2 — FerroShield™ Pro	0.15 ft ³	0.20 ft ³	0.25 ft ³
Stage 3 — ChloroShield™ Pro	0.15 ft ³	0.20 ft ³	0.25 ft ³
Stage 4 — CarbonShield™ Catalytic	1.0 ft ³	1.5 ft ³	2.0 ft ³
Stage 5 — Support underbed	0.25 ft ³	0.35 ft ³	0.50 ft ³
Peak service flow	8 gpm	11 gpm	14 gpm
PFAS-rated flow (EBCT ≥ 90s) ¹	4 gpm	6 gpm	8 gpm
Max. hardness	75 gpg	75 gpg	75 gpg
Mineral tank	10" × 54"	12" × 52"	13" × 54"
Inlet / outlet connection	1"	1"	1"
Approx. shipping weight	175 lb	260 lb	330 lb

¹PFAS reduction depends on empty-bed contact time (EBCT ≥ 90 sec) and influent water chemistry. Size to the PFAS-rated flow for maximum reduction. Larger and commercial configurations available on request.

CARBONSHIELD™

Base	Coconut carbon, MO/Fe-enhanced
Appearance	Greenish-black
Mesh	8×30 · 0.6–2.4 mm
Surface (BET)	2,000–2,500 m ² /g
Bulk density	630–640 kg/m ³
Service life	2–5 yrs (regen.)*

FERROSHIELD /
CHLOROSHIELD

Base	High-purity Cu-Zn
Appearance	Gray metallic
Mechanism	Redox (electron exch.)
Reduces	Cl ₂ · Fe · H ₂ S · metals
Microbial	Bacteriostatic†
Service life	5–8 yrs

SCALESHIELD™

Base	Ceramic crystal beads
Appearance	White / opaque
Mesh	20×35 · 0.55–0.75 mm
Bulk density	780 kg/m ³
Regeneration	None — salt-free
Service life	5–7 yrs

OPERATING PARAMETERS

Service flow direction	Down-flow
Operating temperature	40–100 °F
Operating pressure	25–80 psi
Inlet pH range	6.5–9.5
Max. TDS / salinity	35,000 ppm
Electrical	None · opt. meter

INCLUDED & PACKAGING

- Premium full stainless-steel jacketed mineral tank
- 5-stage proprietary media set (ScaleShield™ + FerroShield™ + ChloroShield™ + CarbonShield™ + underbed)
- Professional Series in/out control head (meter optional)
- Riser, distributor & graded support underbed
- Installation & owner's manual



CERTIFICATIONS & NOTES

ScaleShield™ crystal media is **certified to NSF / ANSI / CAN 61 & 372** (IAPMO). FerroShield™ and ChloroShield™ redox media meet EPA/FDA potable-water standards. *CarbonShield™ is regenerable on-site. †FerroShield™ / ChloroShield™ are bacteriostatic — not a disinfection system. Do not use on water that is microbiologically unsafe without adequate disinfection.

Disclaimer: Information is based on media-manufacturer data, provided in good faith without warranty. Actual contaminant reduction varies with water chemistry, flow rate, contact time and concentration. Product name and model numbers are working placeholders for this draft; specifications subject to change.